ELEC2134 Laboratory 2

Power Engineering Experiment

Overview

Recall that, for maximum power transferred from a source with a finite internal impedance to a load, the impedance of the load must be equal to the complex conjugate of the source impedance, according to the maximum power transfer theorem. Thus, if the source reactance is inductive, the load reactance should be capacitive of equal magnitude and vice versa.

In power systems, industry loads are commonly inductive, consisting of motors, compressors, small-scale transformers, etc., which have low lagging power factors. In order to maintain the power factor **close to unity** (why?), shunt compensators, which are capacitor banks, can be installed throughout the network as needed to improve the power factor.

In this lab, you will validate the condition for maximum power transfer using an AC network at different load conditions and will implement power factor correction using a variable shunt capacitor in the given AC network.

In Australia, most industry operates at the Australian power frequency of 50Hz. In this lab, to simulate these conditions on a breadboard, we will be using 5000Hz, as components required to simulate this at 50Hz are not suited for a breadboard (why?).

Aims

- Verify the maximum power transfer theorem for AC circuits by adjusting the load impedance and observing the power delivered to the load.
- Investigate the relationship between power factor and power transfer.
- Implement power factor correction and demonstrate improvement toward unity power factor.

Prelab Preparation

Complete the following questions before your lab session. They are based on the circuit in Figure 2.1, which uses the same source as the circuits in Section 1 (Figure 2.2) and Section 2 (Figure 2.3) of the Lab Exercises: a 5 V RMS source at f = 5000 Hz with a source impedance made up of the 56 Ω resistor in series with the 100 nF capacitor, connected to a load impedance Z_L.

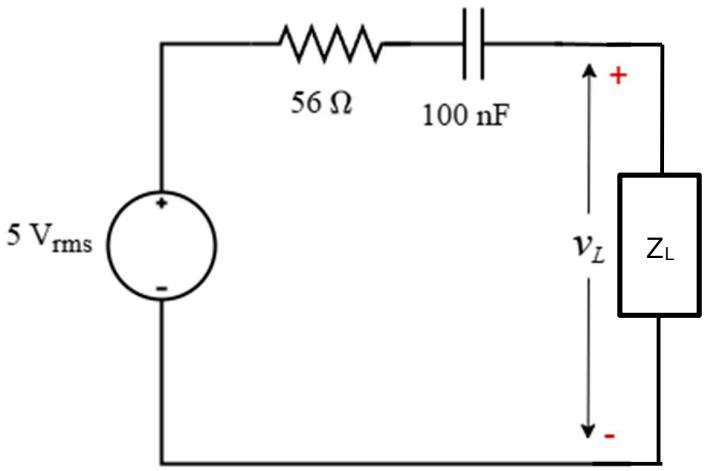


Figure 2.1. Prelab circuit for maximum power transfer and power factor correction.

1. Find the value of Z_L that guarantees maximum average power transferred to the load in the two situations listed below, (i) and (ii). Then, find the corresponding average power.

   i. Z_L is a complex load.

   ii. Z_L is a resistive load.

2. Calculate the power factor for the load impedances (Z_L) in question 1 (i) and (ii).

3. For the scenario mentioned in question 1(i) (i.e., Z_L is a complex load that maximises average power transferred),

   i. If we want to correct the power factor to be 0.99, how can this be achieved?

   ii. What is the value of the component needed to make this happen and what is the average power now, *assuming that the load voltage does not change*[1]?

   iii. What is the benefit of a larger power factor?

4. Use AC analysis simulation to show the reactive power and power factor before and after power factor correction in question 4.

Once you have completed all the questions from this checkpoint, please submit a marking request here: https://forms.cloud.microsoft/r/9mnwX47BP1.

Checkpoint 2.1	Assessor	Marks	Date

[1] In practice, load voltage will change.

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Lab Exercises

Section 1. Maximum power transfer

In this section you will validate the condition for maximum power transfer using an AC network at different load conditions, as per Figure 2.2.

1. Set the signal generator to produce a sinusoidal wave at a frequency of 5000Hz. Adjust the amplitude dial to set the voltage to 5V RMS, which you can determine by using either the oscilloscope's measure function, or the digital multimeter's AC voltage measurement setting.

2. Construct the remainder of the circuit shown in Figure 2.2, using a load resistance $R_L = 10\,\Omega$, which is less than the total resistance of the source. Use a test resistor $R_{test} = 1\,\Omega$.

As oscilloscopes cannot measure current, test resistors can be used in AC circuits to measure phase by converting the current into a measurable voltage to then determine the phase angle. In this case, we have added a very small test resistor R_{test} of value 1Ω in series with the inductor to be able to know the inductor's current, which we know is proportional to the voltage measured in the test resistor $\left(i = \frac{v_{R_{test}}}{1}\right)$. The phase of this voltage $v_{R_{test}}$ is then the same as the phase of the current, so it can then be compared to the phase of the voltage across the entire load, providing the phase angle of the load.

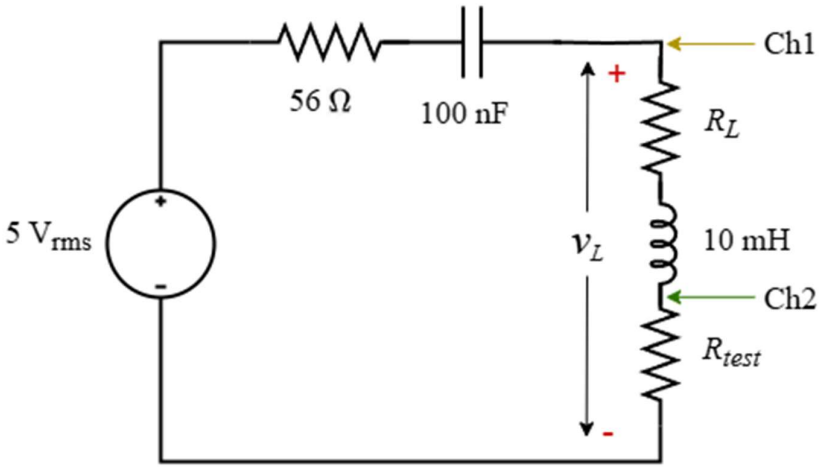


Figure 2.2. Circuit diagram for verification of maximum power transfer theorem.

3. By placing the oscilloscope Channel 1 and Channel 2 probes as shown in Figure 2.2, record the RMS voltage across the load V_L, RMS load current I_L, and load phase θ_L and use these to compute load power P_L for the values of R_L indicated in Table 2.1.

Note that current I_L can be measured using the multimeter set to AC current. Also, in the arrangement above, as previously explained, the phase difference between CH1 and CH2 is θ_L, which can be found using the measurement function of the oscilloscope. Finally, efficiency η is given by:

$$\eta = \frac{Output\ Power}{Input\ Power} = \frac{P_L}{I_L^2 \times R_{total}}$$

Table 2.1. Verification of maximum power transfer theorem.

R_L (Ω)	V_L (V)	I_L (mA)	θ_L (°)	P_L (W)	Efficiency, η

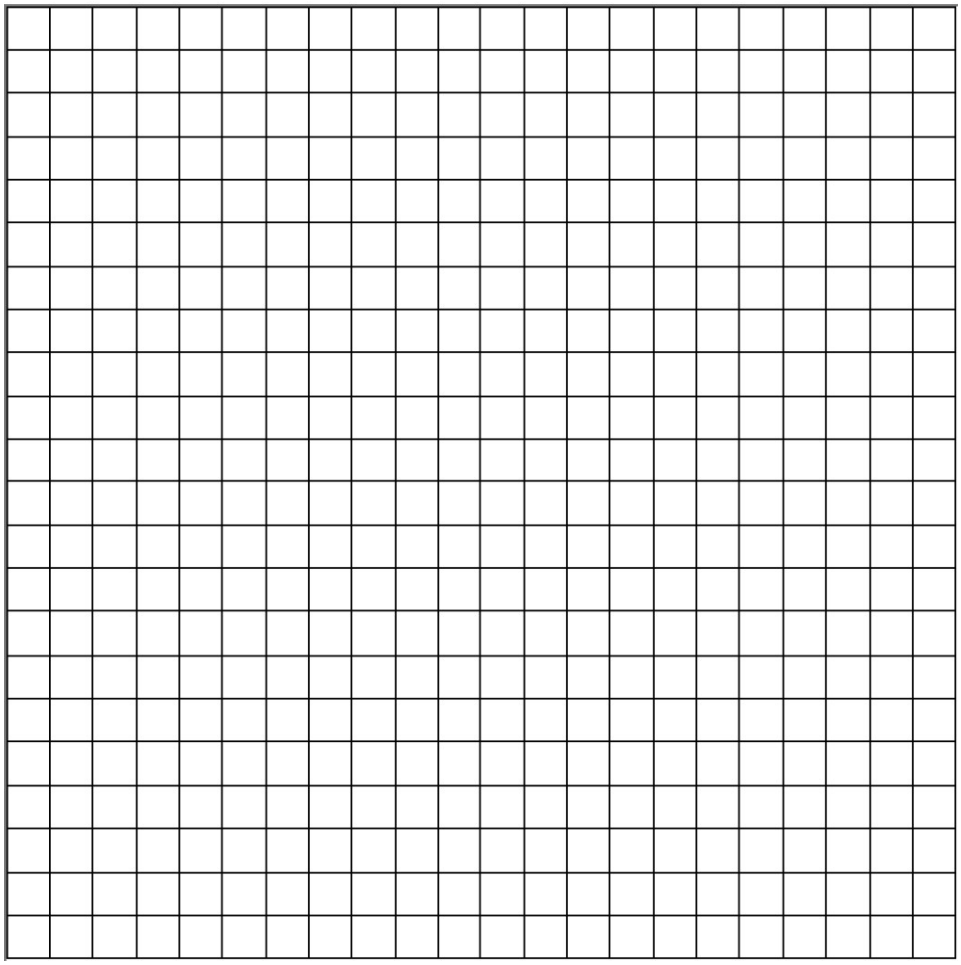
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10					
39					
68					
100					
150					
220					

4. Plot power P_L versus R_L. You are encouraged to practice using Excel, MATLAB, or another graphing software, or you may use the graph paper provided below. R_L should be along the x-axis and P_L should be along the y-axis.

i. From your plot, find the resistance R_L for which the maximum power occurs.

ii. What is the theoretical value of R_L at which maximum power transfer occurs? Is it the same as what you have found from the plot? Compare and comment on your results.



5. On the same graph, plot the efficiency η of against R_L. Where does maximum efficiency occur? How does this compare to maximum power transfer?

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6. Calculate the theoretical value of P_L under ideal condition of maximum power and compare with the measured value. Does it agree with the measured value? Why/why not?

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Checkpoint 2.2	Assessor	Marks	Date

Section 2. Power Factor Correction

For this part, we will connect several values of capacitor C in parallel with the load circuit, as indicated in Figure 2.3. Note that R_L will remain unchanged for the remaining parts of this experiment (R_L =100 Ω).

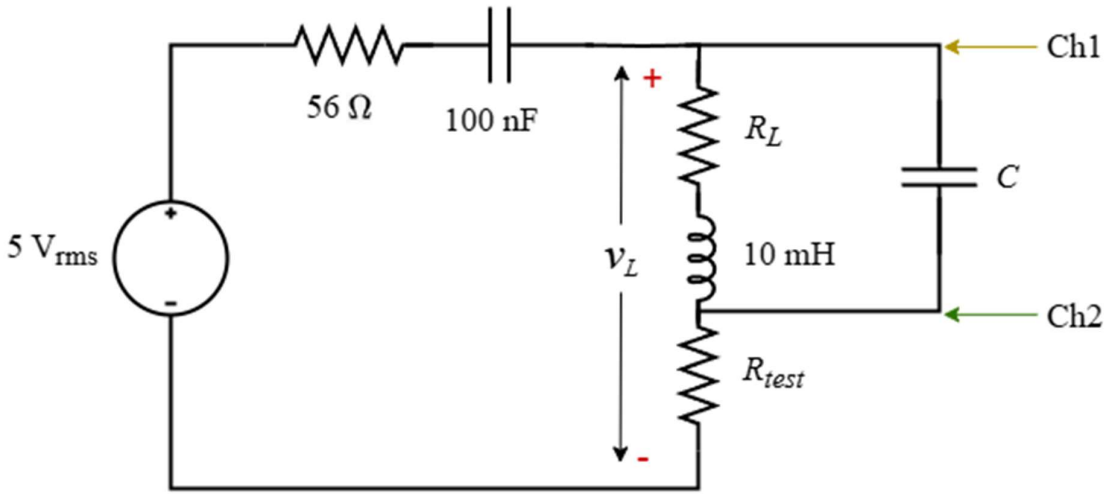


Figure 2.3. Circuit diagram for power factor correction.

1. Copy your results from the relevant row of Table 2.1 (R_L =100 Ω) from Section 1 into the first row of Table 2.2 below and use them to compute the power factor for this row. This result is the one when there is no capacitor C connected.

2. Record the RMS voltage across the load V_L, RMS load current I_L, and load phase θ_L and use these to compute load power P_L and power factor for the values of C indicated in Table 2.2 below.

3. Compare the power factors found when there is no capacitor connected to those with larger capacitors attached. Comment on your results. Explain why the power factor varies.

Table 2.2. Power factor correction.

C (nF)	V_L (V)	I_L (A)	θ_L (°)	P_L (W)	PF

No capacitor					
18					
68					
100					

Once you have completed all the questions from this checkpoint, please submit a marking request here: https://forms.cloud.microsoft/r/9mnwX47BP1.

Checkpoint 2.3	Assessor	Marks	Date